Import Libraries

```
import numpy as np
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense
```

Load and Preprocess Dataset

```
data = """machine learning models learn patterns from data.
sequence models process data step by step.
recurrent neural networks are designed for sequential tasks.
rnn models maintain hidden states across time steps.

long short term memory networks solve long dependency problems.
lstm uses gates to control information flow.
gru models simplify the lstm architecture.
sequence prediction is useful in many applications.

language modeling predicts the next word in a sentence.
speech recognition processes audio sequences.
time series forecasting predicts future values.
music generation creates new melodies.

generative models learn probability distributions.
they generate new samples similar to training data.
sequence generation is widely used in artificial intelligence.
deep learning improves sequence modeling performance."""
```

Tokenization and Numerical Representation

```
tokenizer = Tokenizer()
tokenizer.fit_on_texts([data])

total_words = len(tokenizer.word_index) + 1
print("Total words:", total_words)
```

```
Total words: 87
```

Create Input-Output Sequence Pairs

```
input_sequences = []

for line in data.split('\n'):
    token_list = tokenizer.texts_to_sequences([line])[0]
    for i in range(1, len(token_list)):
        n_gram_seq = token_list[:i+1]
        input_sequences.append(n_gram_seq)

max_seq_len = max(len(seq) for seq in input_sequences)

input_sequences = pad_sequences(input_sequences, maxlen=max_seq_len,
padding='pre')

X = input_sequences[:, :-1]
y = input_sequences[:, -1]

y = tf.keras.utils.to_categorical(y, num_classes=total_words)
```

Build LSTM Model

```
model = Sequential([
    Embedding(total_words, 64, input_length=max_seq_len-1),
    LSTM(100),
    Dense(total_words, activation='softmax')
])

model.compile(loss='categorical_crossentropy', optimizer='adam',
metrics=['accuracy'])
model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
embedding.py:100: UserWarning: Argument `input_length` is deprecated.
Just remove it.
  warnings.warn(
```

```
Model: "sequential"

┏━━━
```

```
│ lstm (LSTM)                     │ ?                      │   0
(unbuilt) │
├─────────────────────────────────┼────────────────────────┼──────────
──────┤
│ dense (Dense)                   │ ?                      │   0
(unbuilt) │
└─────────────────────────────────┴────────────────────────┴──────────
──────┘

 Total params: 0 (0.00 B)

 Trainable params: 0 (0.00 B)

 Non-trainable params: 0 (0.00 B)
```

Train the Model

```
model.fit(X, y, epochs=200, verbose=1)
```

```
Epoch 1/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 6s 18ms/step - accuracy: 0.0105 - loss:
4.4672
Epoch 2/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0526 - loss:
4.4569
Epoch 3/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0526 - loss:
4.4481
Epoch 4/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0526 - loss:
4.4362
Epoch 5/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.0526 - loss:
4.4205
Epoch 6/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0526 - loss:
4.3955
Epoch 7/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.0526 - loss:
4.3576
Epoch 8/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.0526 - loss:
4.2962
Epoch 9/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0526 - loss:
4.2489
Epoch 10/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.0526 - loss:
```

```
4.2391
Epoch 11/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0632 - loss:
4.2098
Epoch 12/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.0632 - loss:
4.1731
Epoch 13/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.0737 - loss:
4.1538
Epoch 14/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.0842 - loss:
4.1273
Epoch 15/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.0947 - loss:
4.0968
Epoch 16/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0947 - loss:
4.0721
Epoch 17/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.0947 - loss:
4.0391
Epoch 18/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.1158 - loss:
3.9999
Epoch 19/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.1263 - loss:
3.9614
Epoch 20/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.1263 - loss:
3.9173
Epoch 21/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.1158 - loss:
3.8711
Epoch 22/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.1158 - loss:
3.8149
Epoch 23/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1158 - loss:
3.7598
Epoch 24/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1158 - loss:
3.6928
Epoch 25/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1263 - loss:
3.6367
Epoch 26/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1263 - loss:
3.5614
```

```
Epoch 27/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1474 - loss:
3.4979
Epoch 28/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1684 - loss:
3.4269
Epoch 29/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.2000 - loss:
3.3521
Epoch 30/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.1895 - loss:
3.2952
Epoch 31/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.2211 - loss:
3.2097
Epoch 32/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.2105 - loss:
3.1152
Epoch 33/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.2947 - loss:
3.0390
Epoch 34/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.3053 - loss:
2.9596
Epoch 35/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.3263 - loss:
2.8609
Epoch 36/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.4105 - loss:
2.8005
Epoch 37/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.4211 - loss:
2.7039
Epoch 38/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.4421 - loss:
2.6059
Epoch 39/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.4526 - loss:
2.5399
Epoch 40/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.5158 - loss:
2.4724
Epoch 41/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.5158 - loss:
2.3976
Epoch 42/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.5263 - loss:
2.3129
Epoch 43/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.5579 - loss:
2.2281
Epoch 44/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6105 - loss:
2.1571
Epoch 45/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.5789 - loss:
2.1137
Epoch 46/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6000 - loss:
2.0327
Epoch 47/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6316 - loss:
1.9864
Epoch 48/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.5895 - loss:
1.9403
Epoch 49/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6211 - loss:
1.8818
Epoch 50/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.6947 - loss:
1.8125
Epoch 51/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6632 - loss:
1.7521
Epoch 52/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6842 - loss:
1.7038
Epoch 53/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.6947 - loss:
1.6597
Epoch 54/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7263 - loss:
1.6191
Epoch 55/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.6947 - loss:
1.5739
Epoch 56/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7368 - loss:
1.5342
Epoch 57/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.7368 - loss:
1.4883
Epoch 58/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.7368 - loss:
1.4465
Epoch 59/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7579 - loss:
```

```
1.4196
Epoch 60/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.7789 - loss:
1.3729
Epoch 61/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7789 - loss:
1.3383
Epoch 62/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.7789 - loss:
1.3168
Epoch 63/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7895 - loss:
1.2814
Epoch 64/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.7895 - loss:
1.2503
Epoch 65/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8000 - loss:
1.2092
Epoch 66/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8000 - loss:
1.1873
Epoch 67/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7789 - loss:
1.1657
Epoch 68/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8105 - loss:
1.1405
Epoch 69/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8105 - loss:
1.1145
Epoch 70/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8105 - loss:
1.0826
Epoch 71/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8105 - loss:
1.0651
Epoch 72/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8211 - loss:
1.0425
Epoch 73/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8105 - loss:
1.0149
Epoch 74/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8211 - loss:
0.9880
Epoch 75/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.8211 - loss:
0.9761
```

```
Epoch 76/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.8421 - loss:
0.9454
Epoch 77/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.8526 - loss:
0.9273
Epoch 78/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8632 - loss:
0.9061
Epoch 79/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8632 - loss:
0.8931
Epoch 80/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8842 - loss:
0.8705
Epoch 81/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.8842 - loss:
0.8588
Epoch 82/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8947 - loss:
0.8370
Epoch 83/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8947 - loss:
0.8154
Epoch 84/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8947 - loss:
0.8049
Epoch 85/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9053 - loss:
0.7823
Epoch 86/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8947 - loss:
0.7693
Epoch 87/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9158 - loss:
0.7533
Epoch 88/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9053 - loss:
0.7417
Epoch 89/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9053 - loss:
0.7251
Epoch 90/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9158 - loss:
0.7094
Epoch 91/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9158 - loss:
0.6942
Epoch 92/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9158 - loss:
0.6793
Epoch 93/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9158 - loss:
0.6661
Epoch 94/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9263 - loss:
0.6551
Epoch 95/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9263 - loss:
0.6432
Epoch 96/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9368 - loss:
0.6305
Epoch 97/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9263 - loss:
0.6171
Epoch 98/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9368 - loss:
0.6042
Epoch 99/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9368 - loss:
0.5959
Epoch 100/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9474 - loss:
0.5836
Epoch 101/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9474 - loss:
0.5706
Epoch 102/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9368 - loss:
0.5581
Epoch 103/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9368 - loss:
0.5482
Epoch 104/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9579 - loss:
0.5378
Epoch 105/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9579 - loss:
0.5312
Epoch 106/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9474 - loss:
0.5211
Epoch 107/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9579 - loss:
0.5125
Epoch 108/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9474 - loss:
```

```
0.4998
Epoch 109/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9579 - loss:
0.4922
Epoch 110/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9579 - loss:
0.4807
Epoch 111/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9579 - loss:
0.4736
Epoch 112/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.4630
Epoch 113/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.4550
Epoch 114/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.4468
Epoch 115/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9579 - loss:
0.4409
Epoch 116/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.4283
Epoch 117/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.4237
Epoch 118/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.4149
Epoch 119/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.4063
Epoch 120/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.3980
Epoch 121/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.3913
Epoch 122/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9684 - loss:
0.3843
Epoch 123/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.3761
Epoch 124/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.3710
```

```
Epoch 125/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9579 - loss:
0.3644
Epoch 126/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9684 - loss:
0.3586
Epoch 127/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.3517
Epoch 128/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.9684 - loss:
0.3474
Epoch 129/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.9684 - loss:
0.3432
Epoch 130/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 0.9684 - loss:
0.3365
Epoch 131/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9684 - loss:
0.3307
Epoch 132/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 0.9684 - loss:
0.3240
Epoch 133/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9684 - loss:
0.3179
Epoch 134/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9684 - loss:
0.3110
Epoch 135/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.9684 - loss:
0.3083
Epoch 136/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9684 - loss:
0.3024
Epoch 137/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9579 - loss:
0.2971
Epoch 138/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9684 - loss:
0.2925
Epoch 139/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9684 - loss:
0.2871
Epoch 140/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9684 - loss:
0.2839
Epoch 141/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.9684 - loss:
0.2774
Epoch 142/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.9684 - loss:
0.2731
Epoch 143/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - accuracy: 0.9684 - loss:
0.2707
Epoch 144/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 0.9684 - loss:
0.2661
Epoch 145/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9789 - loss:
0.2618
Epoch 146/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - accuracy: 0.9579 - loss:
0.2576
Epoch 147/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9579 - loss:
0.2535
Epoch 148/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.2485
Epoch 149/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2452
Epoch 150/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2409
Epoch 151/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2388
Epoch 152/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.2349
Epoch 153/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.2313
Epoch 154/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.2276
Epoch 155/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.2242
Epoch 156/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2207
Epoch 157/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
```

```
0.2188
Epoch 158/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2146
Epoch 159/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.2127
Epoch 160/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2101
Epoch 161/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9789 - loss:
0.2075
Epoch 162/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.2042
Epoch 163/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.2009
Epoch 164/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.1981
Epoch 165/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1969
Epoch 166/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1938
Epoch 167/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1925
Epoch 168/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1887
Epoch 169/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1866
Epoch 170/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1837
Epoch 171/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.1820
Epoch 172/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1798
Epoch 173/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.1769
Epoch 174/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1754
Epoch 175/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1721
Epoch 176/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.1706
Epoch 177/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9789 - loss:
0.1688
Epoch 178/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1674
Epoch 179/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1652
Epoch 180/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1624
Epoch 181/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1612
Epoch 182/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.1589
Epoch 183/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1576
Epoch 184/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1557
Epoch 185/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.1546
Epoch 186/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.1536
Epoch 187/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1503
Epoch 188/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1488
Epoch 189/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1489
Epoch 190/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
```

```
0.1466
Epoch 191/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9684 - loss:
0.1441
Epoch 192/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1438
Epoch 193/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.1418
Epoch 194/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1395
Epoch 195/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1388
Epoch 196/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1370
Epoch 197/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.1359
Epoch 198/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1351
Epoch 199/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.1328
Epoch 200/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.1317

<keras.src.callbacks.history.History at 0x78afa8f568d0>
```

Generate New Sequences

```
def generate_text(seed_text, next_words):
    for _ in range(next_words):
        token_list = tokenizer.texts_to_sequences([seed_text])[0]
        token_list = pad_sequences([token_list], maxlen=max_seq_len-1,
padding='pre')
        predicted = np.argmax(model.predict(token_list), axis=-1)[0]

        for word, index in tokenizer.word_index.items():
            if index == predicted:
                seed_text += " " + word
                break
    return seed_text
```

```
print(generate_text("machine learning", 5))
print(generate_text("sequence models", 5))
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 176ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
machine learning models learn patterns from data
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
sequence models process data step by step
```

Component II: Transformer Based Sequence Generation

Tokenization

```
import numpy as np
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.layers import Embedding, Dense,
LayerNormalization, Dropout, MultiHeadAttention,
GlobalAveragePooling1D, Input
from tensorflow.keras.models import Model

tokenizer = Tokenizer()
tokenizer.fit_on_texts(data.split('\n'))

total_words = len(tokenizer.word_index) + 1
```

Positional Encoding

```
input_sequences = []

for line in data.split('\n'):
    token_list = tokenizer.texts_to_sequences([line])[0]
    for i in range(1, len(token_list)):
```

```
        input_sequences.append(token_list[:i+1])

max_len = max(len(seq) for seq in input_sequences)

input_sequences = pad_sequences(input_sequences, maxlen=max_len,
padding='pre')

X = input_sequences[:, :-1]
y = input_sequences[:, -1]
```

Transformer Model

```
def transformer_block(inputs, head_size, num_heads, ff_dim,
dropout=0.1):
    x = MultiHeadAttention(num_heads=num_heads, key_dim=head_size)
(inputs, inputs)
    x = Dropout(dropout)(x)
    x = LayerNormalization(epsilon=1e-6)(x + inputs)

    ffn = Dense(ff_dim, activation="relu")(x)
    ffn = Dense(inputs.shape[-1])(ffn)
    ffn = Dropout(dropout)(ffn)

    return LayerNormalization(epsilon=1e-6)(x + ffn)
```

Build Transformer Model

```
embed_dim = 64
num_heads = 2
ff_dim = 128

inputs = Input(shape=(max_len - 1,))
x = Embedding(total_words, embed_dim)(inputs)

x = transformer_block(x, embed_dim, num_heads, ff_dim)
x = GlobalAveragePooling1D()(x)

outputs = Dense(total_words, activation="softmax")(x)

model = Model(inputs, outputs)
model.compile(loss='sparse_categorical_crossentropy',
optimizer='adam', metrics=['accuracy'])

model.summary()
```

```
Model: "functional_5"
```

```
┏━━━
```

```
├──────────────────────┼──────────────────────┼──────────────┼──────────────────────┤
│ add_3 (Add)          │ (None, 8, 64)        │            0 │ layer_normalizat… │
│                      │                      │              │ dropout_6[0][0]   │
├──────────────────────┼──────────────────────┼──────────────┼──────────────────────┤
│ layer_normalizatio… │ (None, 8, 64)        │          128 │ add_3[0][0]       │
│ (LayerNormalizatio… │                      │              │                   │
├──────────────────────┼──────────────────────┼──────────────┼──────────────────────┤
│ global_average_poo… │ (None, 64)           │            0 │ layer_normalizat… │
│ (GlobalAveragePool… │                      │              │                   │
├──────────────────────┼──────────────────────┼──────────────┼──────────────────────┤
│ dense_9 (Dense)      │ (None, 87)           │        5,655 │ global_average_p… │
└──────────────────────┴──────────────────────┴──────────────┴──────────────────────┘

 Total params: 61,271 (239.34 KB)

 Trainable params: 61,271 (239.34 KB)

 Non-trainable params: 0 (0.00 B)
```

Prepare Data for Transformer

```
model.fit(X, y, epochs=200, verbose=1)
```

```
Epoch 1/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 10s 2s/step - accuracy: 0.0000e+00 - loss: 4.5987
Epoch 2/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.0632 - loss: 4.2698
Epoch 3/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.1053 - loss: 4.0777
Epoch 4/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.1053 - loss: 3.9521
Epoch 5/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.1579 - loss:
3.8334
Epoch 6/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.1579 - loss:
3.7132
Epoch 7/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.2000 - loss:
3.5843
Epoch 8/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.2105 - loss:
3.4828
Epoch 9/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.2000 - loss:
3.4023
Epoch 10/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.2632 - loss:
3.3106
Epoch 11/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.3158 - loss:
3.2135
Epoch 12/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.3158 - loss:
3.1308
Epoch 13/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.3263 - loss:
3.0376
Epoch 14/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.3158 - loss:
2.9428
Epoch 15/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.3368 - loss:
2.8564
Epoch 16/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.3789 - loss:
2.7486
Epoch 17/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.4316 - loss:
2.6275
Epoch 18/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.4105 - loss:
2.5103
Epoch 19/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.4421 - loss:
2.3896
Epoch 20/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.4947 - loss:
2.2696
Epoch 21/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.4105 - loss:
```

```
2.1812
Epoch 22/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.4737 - loss:
2.0996
Epoch 23/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.5789 - loss:
1.9814
Epoch 24/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.6316 - loss:
1.8714
Epoch 25/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.6421 - loss:
1.7767
Epoch 26/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.6211 - loss:
1.7241
Epoch 27/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.7263 - loss:
1.6553
Epoch 28/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.7579 - loss:
1.5597
Epoch 29/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.7684 - loss:
1.4732
Epoch 30/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7789 - loss:
1.4327
Epoch 31/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8421 - loss:
1.3749
Epoch 32/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8947 - loss:
1.2723
Epoch 33/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8737 - loss:
1.2325
Epoch 34/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8842 - loss:
1.1928
Epoch 35/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9474 - loss:
1.1056
Epoch 36/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8947 - loss:
1.0649
Epoch 37/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9368 - loss:
1.0154
```

```
Epoch 38/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9579 - loss:
0.9632
Epoch 39/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9368 - loss:
0.9122
Epoch 40/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9474 - loss:
0.8695
Epoch 41/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9368 - loss:
0.8400
Epoch 42/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9579 - loss:
0.8207
Epoch 43/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.7592
Epoch 44/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9579 - loss:
0.7367
Epoch 45/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9579 - loss:
0.7009
Epoch 46/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.6538
Epoch 47/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9474 - loss:
0.6311
Epoch 48/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9579 - loss:
0.6064
Epoch 49/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.5813
Epoch 50/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.5437
Epoch 51/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.5181
Epoch 52/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.4935
Epoch 53/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.4640
Epoch 54/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.4511
Epoch 55/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.4439
Epoch 56/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9579 - loss:
0.4103
Epoch 57/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.3901
Epoch 58/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.3827
Epoch 59/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.3580
Epoch 60/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.3436
Epoch 61/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.3399
Epoch 62/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.3262
Epoch 63/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.3106
Epoch 64/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.3022
Epoch 65/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.2881
Epoch 66/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.2847
Epoch 67/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.2766
Epoch 68/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.2589
Epoch 69/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.2458
Epoch 70/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
```

```
0.2364
Epoch 71/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.2334
Epoch 72/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.2371
Epoch 73/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.2245
Epoch 74/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.2192
Epoch 75/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.2153
Epoch 76/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.2097
Epoch 77/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.2070
Epoch 78/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.2177
Epoch 79/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9895 - loss:
0.1880
Epoch 80/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1993
Epoch 81/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.1827
Epoch 82/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1780
Epoch 83/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.1691
Epoch 84/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.1679
Epoch 85/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1675
Epoch 86/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.1708
```

```
Epoch 87/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.1527
Epoch 88/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.1606
Epoch 89/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1592
Epoch 90/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.1481
Epoch 91/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9895 - loss:
0.1408
Epoch 92/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.1415
Epoch 93/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.1364
Epoch 94/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9895 - loss:
0.1297
Epoch 95/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.1355
Epoch 96/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.1335
Epoch 97/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1301
Epoch 98/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1278
Epoch 99/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.1183
Epoch 100/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9895 - loss:
0.1164
Epoch 101/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.1169
Epoch 102/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.1106
Epoch 103/200
```

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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.1127
Epoch 104/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9895 - loss:
0.1053
Epoch 105/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1103
Epoch 106/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.1012
Epoch 107/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1032
Epoch 108/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.0974
Epoch 109/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0972
Epoch 110/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1038
Epoch 111/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0992
Epoch 112/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.1023
Epoch 113/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0931
Epoch 114/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0980
Epoch 115/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0920
Epoch 116/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0976
Epoch 117/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0924
Epoch 118/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9684 - loss:
0.0938
Epoch 119/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
```

```
0.0884
Epoch 120/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.0847
Epoch 121/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9895 - loss:
0.0836
Epoch 122/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0819
Epoch 123/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0836
Epoch 124/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0859
Epoch 125/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0872
Epoch 126/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9789 - loss:
0.0802
Epoch 127/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0857
Epoch 128/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0877
Epoch 129/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0821
Epoch 130/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.0793
Epoch 131/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0756
Epoch 132/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0784
Epoch 133/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0779
Epoch 134/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0757
Epoch 135/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0751
```

```
Epoch 136/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0746
Epoch 137/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.0783
Epoch 138/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0784
Epoch 139/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0745
Epoch 140/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0713
Epoch 141/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0693
Epoch 142/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0745
Epoch 143/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0705
Epoch 144/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0656
Epoch 145/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0700
Epoch 146/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0708
Epoch 147/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0661
Epoch 148/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0669
Epoch 149/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9895 - loss:
0.0668
Epoch 150/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9684 - loss:
0.0691
Epoch 151/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9789 - loss:
0.0691
Epoch 152/200
```

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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - accuracy: 0.9789 - loss:
0.0712
Epoch 153/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9895 - loss:
0.0681
Epoch 154/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9789 - loss:
0.0627
Epoch 155/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9684 - loss:
0.0628
Epoch 156/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0620
Epoch 157/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9789 - loss:
0.0593
Epoch 158/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.9684 - loss:
0.0637
Epoch 159/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9684 - loss:
0.0642
Epoch 160/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0625
Epoch 161/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0620
Epoch 162/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9684 - loss:
0.0619
Epoch 163/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0649
Epoch 164/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9684 - loss:
0.0624
Epoch 165/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0605
Epoch 166/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9684 - loss:
0.0585
Epoch 167/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9895 - loss:
0.0547
Epoch 168/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9684 - loss:
0.0595
```

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Epoch 169/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.0600
Epoch 170/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9789 - loss:
0.0604
Epoch 171/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9684 - loss:
0.0650
Epoch 172/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0599
Epoch 173/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9684 - loss:
0.0579
Epoch 174/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.0609
Epoch 175/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0595
Epoch 176/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9789 - loss:
0.0638
Epoch 177/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9789 - loss:
0.0638
Epoch 178/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9895 - loss:
0.0550
Epoch 179/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9789 - loss:
0.0587
Epoch 180/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9684 - loss:
0.0627
Epoch 181/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9789 - loss:
0.0532
Epoch 182/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.9789 - loss:
0.0597
Epoch 183/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9789 - loss:
0.0625
Epoch 184/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9684 - loss:
0.0580
Epoch 185/200
```

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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0565
Epoch 186/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0574
Epoch 187/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9684 - loss:
0.0568
Epoch 188/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9895 - loss:
0.0515
Epoch 189/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0557
Epoch 190/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0581
Epoch 191/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0519
Epoch 192/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9789 - loss:
0.0555
Epoch 193/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9684 - loss:
0.0560
Epoch 194/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9684 - loss:
0.0533
Epoch 195/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0552
Epoch 196/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9684 - loss:
0.0560
Epoch 197/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9789 - loss:
0.0561
Epoch 198/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9789 - loss:
0.0502
Epoch 199/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9789 - loss:
0.0513
Epoch 200/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.9789 - loss:
0.0563

<keras.src.callbacks.history.History at 0x78af054bea20>
```

Train Transformer Model

```
transformer_model.fit(X_t, y_t, epochs=200, verbose=1)
```

```
Epoch 1/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0345
Epoch 2/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0341
Epoch 3/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0337
Epoch 4/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0333
Epoch 5/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0328
Epoch 6/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0325
Epoch 7/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0321
Epoch 8/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0317
Epoch 9/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0313
Epoch 10/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0310
Epoch 11/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0306
Epoch 12/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0303
Epoch 13/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0299
Epoch 14/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0296
Epoch 15/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0293
```

```
Epoch 16/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0290
Epoch 17/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0287
Epoch 18/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0284
Epoch 19/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0281
Epoch 20/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0278
Epoch 21/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0275
Epoch 22/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0272
Epoch 23/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0269
Epoch 24/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0267
Epoch 25/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0264
Epoch 26/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0261
Epoch 27/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0259
Epoch 28/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0256
Epoch 29/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0254
Epoch 30/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0251
Epoch 31/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0249
Epoch 32/200
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0247
Epoch 33/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0244
Epoch 34/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0242
Epoch 35/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0240
Epoch 36/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0237
Epoch 37/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0235
Epoch 38/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0233
Epoch 39/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0231
Epoch 40/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0229
Epoch 41/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0227
Epoch 42/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 60ms/step - accuracy: 1.0000 - loss:
0.0225
Epoch 43/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step - accuracy: 1.0000 - loss:
0.0223
Epoch 44/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 54ms/step - accuracy: 1.0000 - loss:
0.0221
Epoch 45/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0219
Epoch 46/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0217
Epoch 47/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0215
Epoch 48/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
```

```
0.0214
Epoch 49/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0212
Epoch 50/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0210
Epoch 51/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0208
Epoch 52/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0206
Epoch 53/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0205
Epoch 54/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0203
Epoch 55/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0201
Epoch 56/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0200
Epoch 57/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0198
Epoch 58/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0197
Epoch 59/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0195
Epoch 60/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0193
Epoch 61/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0192
Epoch 62/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0190
Epoch 63/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 58ms/step - accuracy: 1.0000 - loss:
0.0189
Epoch 64/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0187
```

```
Epoch 65/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0186
Epoch 66/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0185
Epoch 67/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0183
Epoch 68/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0182
Epoch 69/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0180
Epoch 70/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0179
Epoch 71/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0178
Epoch 72/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0176
Epoch 73/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0175
Epoch 74/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0174
Epoch 75/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0172
Epoch 76/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0171
Epoch 77/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0170
Epoch 78/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0169
Epoch 79/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0167
Epoch 80/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0166
Epoch 81/200
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0165
Epoch 82/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0164
Epoch 83/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0163
Epoch 84/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 56ms/step - accuracy: 1.0000 - loss:
0.0162
Epoch 85/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 52ms/step - accuracy: 1.0000 - loss:
0.0160
Epoch 86/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 62ms/step - accuracy: 1.0000 - loss:
0.0159
Epoch 87/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0158
Epoch 88/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0157
Epoch 89/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0156
Epoch 90/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0155
Epoch 91/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0154
Epoch 92/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0153
Epoch 93/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0152
Epoch 94/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0151
Epoch 95/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0150
Epoch 96/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0149
Epoch 97/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
```

```
0.0148
Epoch 98/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0147
Epoch 99/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0146
Epoch 100/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0145
Epoch 101/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0144
Epoch 102/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step - accuracy: 1.0000 - loss:
0.0143
Epoch 103/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0142
Epoch 104/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 54ms/step - accuracy: 1.0000 - loss:
0.0141
Epoch 105/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0140
Epoch 106/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0139
Epoch 107/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0138
Epoch 108/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0137
Epoch 109/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0137
Epoch 110/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0136
Epoch 111/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0135
Epoch 112/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0134
Epoch 113/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0133
```

```
Epoch 114/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0132
Epoch 115/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0132
Epoch 116/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0131
Epoch 117/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0130
Epoch 118/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0129
Epoch 119/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss:
0.0128
Epoch 120/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0128
Epoch 121/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0127
Epoch 122/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0126
Epoch 123/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0125
Epoch 124/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0124
Epoch 125/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0124
Epoch 126/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 58ms/step - accuracy: 1.0000 - loss:
0.0123
Epoch 127/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0122
Epoch 128/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0122
Epoch 129/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0121
Epoch 130/200
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0120
Epoch 131/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0119
Epoch 132/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0119
Epoch 133/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0118
Epoch 134/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0117
Epoch 135/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0117
Epoch 136/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0116
Epoch 137/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0115
Epoch 138/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0115
Epoch 139/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0114
Epoch 140/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0113
Epoch 141/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0113
Epoch 142/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0112
Epoch 143/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0111
Epoch 144/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0111
Epoch 145/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0110
Epoch 146/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 53ms/step - accuracy: 1.0000 - loss:
```

```
0.0110
Epoch 147/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 55ms/step - accuracy: 1.0000 - loss:
0.0109
Epoch 148/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0108
Epoch 149/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 1.0000 - loss:
0.0108
Epoch 150/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0107
Epoch 151/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0107
Epoch 152/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0106
Epoch 153/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step - accuracy: 1.0000 - loss:
0.0105
Epoch 154/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0105
Epoch 155/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0104
Epoch 156/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step - accuracy: 1.0000 - loss:
0.0104
Epoch 157/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0103
Epoch 158/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0103
Epoch 159/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0102
Epoch 160/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0102
Epoch 161/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0101
Epoch 162/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0100
```

```
Epoch 163/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0100
Epoch 164/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0099
Epoch 165/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0099
Epoch 166/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0098
Epoch 167/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 53ms/step - accuracy: 1.0000 - loss:
0.0098
Epoch 168/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step - accuracy: 1.0000 - loss:
0.0097
Epoch 169/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0097
Epoch 170/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0096
Epoch 171/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0096
Epoch 172/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0095
Epoch 173/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0095
Epoch 174/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - accuracy: 1.0000 - loss:
0.0094
Epoch 175/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 53ms/step - accuracy: 1.0000 - loss:
0.0094
Epoch 176/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0093
Epoch 177/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - accuracy: 1.0000 - loss:
0.0093
Epoch 178/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0093
Epoch 179/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
```

```
0.0092
Epoch 180/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - accuracy: 1.0000 - loss:
0.0092
Epoch 181/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - loss:
0.0091
Epoch 182/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step - accuracy: 1.0000 - loss:
0.0091
Epoch 183/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 57ms/step - accuracy: 1.0000 - loss:
0.0090
Epoch 184/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 145ms/step - accuracy: 1.0000 - loss:
0.0090
Epoch 185/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 85ms/step - accuracy: 1.0000 - loss:
0.0089
Epoch 186/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 69ms/step - accuracy: 1.0000 - loss:
0.0089
Epoch 187/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 69ms/step - accuracy: 1.0000 - loss:
0.0088
Epoch 188/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 64ms/step - accuracy: 1.0000 - loss:
0.0088
Epoch 189/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 65ms/step - accuracy: 1.0000 - loss:
0.0088
Epoch 190/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 65ms/step - accuracy: 1.0000 - loss:
0.0087
Epoch 191/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 145ms/step - accuracy: 1.0000 - loss:
0.0087
Epoch 192/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 64ms/step - accuracy: 1.0000 - loss:
0.0086
Epoch 193/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 70ms/step - accuracy: 1.0000 - loss:
0.0086
Epoch 194/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 69ms/step - accuracy: 1.0000 - loss:
0.0086
Epoch 195/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 61ms/step - accuracy: 1.0000 - loss:
0.0085
```

```
Epoch 196/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 66ms/step - accuracy: 1.0000 - loss:
0.0085
Epoch 197/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 76ms/step - accuracy: 1.0000 - loss:
0.0084
Epoch 198/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 78ms/step - accuracy: 1.0000 - loss:
0.0084
Epoch 199/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step - accuracy: 1.0000 - loss:
0.0083
Epoch 200/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 62ms/step - accuracy: 1.0000 - loss:
0.0083

<keras.src.callbacks.history.History at 0x78af050ba810>
```

Generate Sequences using Transformer

```
index_to_word = {v: k for k, v in tokenizer.word_index.items()}

def sample_with_temperature(preds, temperature=0.8):
    preds = np.asarray(preds).astype('float64')
    preds = np.log(preds + 1e-9) / temperature
    exp_preds = np.exp(preds)
    preds = exp_preds / np.sum(exp_preds)
    return np.random.choice(len(preds), p=preds)

def generate_text(seed_text, next_words=5, temperature=0.8):
    for _ in range(next_words):
        seq = tokenizer.texts_to_sequences([seed_text])[0]
        seq = pad_sequences([seq], maxlen=max_len - 1, padding='pre')

        preds = model.predict(seq, verbose=0)[0]
        pred_index = sample_with_temperature(preds, temperature)

        if pred_index == 0:
            continue

        next_word = index_to_word.get(pred_index)

        if next_word is None:
            break

        seed_text += " " + next_word

    return seed_text
```

```
print(generate_text("deep learning"))
print(generate_text("sequence models"))
print(generate_text("transformers are"))
```

```
deep learning improves sequence modeling performance performance
sequence models process data step by step
transformers are designed for sequential tasks tasks
```

```
from google.colab import drive
drive.mount('/content/drive')
BASE_DIR = "/content/drive/MyDrive/Colab Notebooks/GenAI/Lab 9/"
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```

```
# Save LSTM model

model.save(BASE_DIR+"lstm_sequence_model.h5")

# Save Transformer model
transformer_model.save(BASE_DIR+"transformer_sequence_model.h5")
```

```
WARNING:absl:You are saving your model as an HDF5 file via
`model.save()` or `keras.saving.save_model(model)`. This file format
is considered legacy. We recommend using instead the native Keras
format, e.g. `model.save('my_model.keras')` or
`keras.saving.save_model(model, 'my_model.keras')`.
WARNING:absl:You are saving your model as an HDF5 file via
`model.save()` or `keras.saving.save_model(model)`. This file format
is considered legacy. We recommend using instead the native Keras
format, e.g. `model.save('my_model.keras')` or
`keras.saving.save_model(model, 'my_model.keras')`.
```

```
import pickle
with open(BASE_DIR+"tokenizer.pkl", "wb") as f:
    pickle.dump(tokenizer, f)
```